Worksheet 6.4 — The Dot Product

Grade 12 Calculus and Vectors (MCV4U) · Unit 6: Geometry and Algebra of Vectors

The Big Ideas

$\vec{a}\cdot\vec{b} = a_1b_1 + a_2b_2(+a_3b_3) = |\vec{a}||\vec{b}|\cos\theta$. Gives the angle; zero means perpendicular; work $= \vec{F}\cdot\vec{d}$.

- Components: $\vec{a}\cdot\vec{b} = a_1b_1 + a_2b_2 + a_3b_3$.
- $\cos\theta = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$.
- Perpendicular $\iff \vec{a}\cdot\vec{b} = 0$; work $= \vec{F}\cdot\vec{d}$.

The Big Picture

Multiplying to a scalar

The **dot product** $\vec{u}\cdot\vec{v} = u_1v_1 + u_2v_2 + \cdots$ produces a *scalar*, not a vector.

The angle connection

It also equals $|\vec{u}|\,|\vec{v}|\cos\theta$, so it measures how aligned two vectors are and gives the angle between them.

Perpendicular test

Two nonzero vectors are perpendicular exactly when their dot product is 0.

Worked Examples

Example 1 — Dot product (2-D)

Find $(2,3)\cdot(4,-1)$.

Solution. **Step 1 (multiply matching components):** $(2)(4)$ and $(3)(-1)$.

Step 2 (add the products): $8 + (-3)$.

Conclusion: $= 5$.

Example 2 — Dot product (3-D)

Find $(1,2,3)\cdot(2,0,-1)$.

Solution. **Step 1 (multiply matching components):** $(1)(2)$, $(2)(0)$, $(3)(-1)$.

Step 2 (add the products): $2 + 0 + (-3)$.

Conclusion: $= -1$.

Example 3 — Angle

Angle between $(1,0)$ and $(1,1)$?

Solution. **Step 1 (find the dot product):** $(1)(1) + (0)(1) = 1$.

Step 2 (find both magnitudes): $|(1,0)| = 1$; $|(1,1)| = \sqrt{2}$.

Step 3 (apply $\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$): $\cos\theta = \dfrac{1}{1\cdot\sqrt{2}} = \dfrac{1}{\sqrt{2}}$.

Conclusion: $\theta = 45°$.

Example 4 — Perpendicular

Are $(2,3)$ and $(3,-2)$ perpendicular?

Solution. **Step 1 (compute the dot product):** $(2)(3)+(3)(-2)=6-6$.

Step 2 (recall the perpendicularity test, dot product $=0$): the result is 0.

Conclusion: yes, perpendicular.

Example 5 — Work

$\vec{F}=(3,4)$ N, $\vec{d}=(2,0)$ m. Find the work.

Solution. **Step 1 (recall the work formula):** $W=\vec{F}\cdot\vec{d}$.

Step 2 (compute the dot product): $(3)(2)+(4)(0)=6+0$.

Conclusion: $W=6$ J.

Example 6 — Dot product

Find $(1,2)\cdot(3,4)$.

Solution. **Step 1 (multiply matching components and add):** $(1)(3)+(2)(4)=3+8$.

Conclusion: $=11$.

Example 7 — Dot product (3-D)

Find $(2,-1,3)\cdot(1,4,2)$.

Solution. **Step 1 (multiply matching components and add):** $(2)(1)+(-1)(4)+(3)(2)=2-4+6$.

Conclusion: $=4$.

Example 8 — Angle

Angle between $(1,0)$ and $(0,1)$?

Solution. **Step 1 (compute the dot product):** $(1)(0)+(0)(1)=0$.

Step 2 (interpret a zero dot product): the vectors are perpendicular.

Conclusion: $90°$.

Example 9 — Work

$\vec{F}=(5,0)$, $\vec{d}=(3,4)$. Find the work.

Solution. **Step 1 (apply $W=\vec{F}\cdot\vec{d}$):** $(5)(3)+(0)(4)=15+0$.

Conclusion: $W=15$ J.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find $(1, 2) \cdot (3, 4)$.

Question 2

Find $(2, -1, 3) \cdot (1, 4, 2)$.

Question 3

Are $(1, 2)$ and $(4, -2)$ perpendicular?

Question 4

Angle between $(1, 0)$ and $(0, 1)$?

Question 5

Work of $\vec{F} = (5, 0)$ along $\vec{d} = (3, 4)$?

Question 6

Find $(3,1)\cdot(2,5)$.

Question 7

Find $(1,1,1)\cdot(2,3,4)$.

Question 8

Are $(3,4)$ and $(4,-3)$ perpendicular?

Question 9

Angle between $(2,0)$ and $(2,2)$?

Question 10

Work of $\vec{F} = (2, 3)$ along $\vec{d} = (4, 0)$?

Question 11

Find $(0, 5) \cdot (5, 0)$.

Question 12

Find $|\vec{a}|$ from $\vec{a} \cdot \vec{a} = 25$.

Question 13 — Challenge

Find the angle between $(3, 4)$ and $(4, 3)$ to the nearest degree.

Question Recap & Answer Key

Question Recap

1. Find $(1,2)\cdot(3,4)$.
2. Find $(2,-1,3)\cdot(1,4,2)$.
3. Are $(1,2)$ and $(4,-2)$ perpendicular?
4. Angle between $(1,0)$ and $(0,1)$?
5. Work of $\vec{F}=(5,0)$ along $\vec{d}=(3,4)$?
6. Find $(3,1)\cdot(2,5)$.
7. Find $(1,1,1)\cdot(2,3,4)$.
8. Are $(3,4)$ and $(4,-3)$ perpendicular?
9. Angle between $(2,0)$ and $(2,2)$?
10. Work of $\vec{F}=(2,3)$ along $\vec{d}=(4,0)$?
11. Find $(0,5)\cdot(5,0)$.
12. Find $|\vec{a}|$ from $\vec{a}\cdot\vec{a}=25$.
13. Find the angle between $(3,4)$ and $(4,3)$ to the nearest degree.

✓ Answer Key

1. 11
2. 4
3. 0: yes
4. 90°
5. 15 J
6. 11
7. 9
8. 0: yes
9. 45°
10. 8 J
11. 0
12. 5
13. $\cos\theta=\frac{24}{25}\Rightarrow\theta\approx16^\circ$